Segmentation-Guided Semantic-Aware Self-Supervised Denoising for SAR Image

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***Abstract*— Synthetic aperture radar (SAR) images often suffer from speckle noise, which can degrade their visual quality and affect downstream applications. Recently, supervised and self-supervised methods have been proposed by virtue of deep learning with synthetic "noisy–clean" image pairs or only real noisy images as training data, respectively. Among these methods, by avoiding artifact problems in real SAR image denoising, self-supervised methods solve the domain gap problem of supervised methods and hence have attracted significant attention. However, existing self-supervised denoising methods essentially rely on pixel information of images and ignore the corresponding semantic information, which makes them challenging to remove speckle noise while retaining the detailed features. To this end, we propose a segmentation-guided semantic-aware self-supervised denoising method for SAR images, namely, SARDeSeg, where a segmentation network is incorporated with a denoising network and guides it to learn and be aware of the semantic information of the input noisy SAR images. In addition, a wavelet transform-based connector is introduced to efficiently transmit semantic information between the denoising network and the segmentation network, together with an edge-aware smoothing loss to improve speckle noise suppression while preserving edge features. The experimental results demonstrate that the proposed SARDeSeg outperforms state-of-the-art (SOTA) denoising methods for SAR images, particularly in preserving detailed edge features.**

***Index Terms*— Edge feature preservation, image denoising, self-supervised learning, speckle noise, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is often employed as a powerful observation tool due to its ability to penetrate through various environmental conditions (e.g., rain, dust, and clouds) and provide all-day and all-weather imaging. As a result, SAR plays a crucial role in numerous industries, e.g., disaster assessment [1], ocean monitoring [2], and change detection [3]. However, accounting for the inherent coherence of the SAR imaging system, speckle noise is a common problem that affects SAR image quality and poses a significant challenge for visual analysis [4]. Therefore, denoising SAR images, which refers to the reduction of speckle noise, is a crucial step in SAR image processing and has been hailed as one of the subjects in extensive research. Numerous traditional methods have been proposed to denoise SAR images, e.g., wavelet-based filters [5], [6], variational-based models [7], [8], [9], and nonlocal-based methods [10], [11].

However, despite their effectiveness in reducing speckle noise, traditional denoising methods have certain limitations. Wavelet-based filters may produce ringing artifacts, visible as oscillatory patterns around sharp edges in the image. Variational-based models may produce oversmoothed results when the regularization term is selected inappropriately. In addition, nonlocal-based methods may introduce structured signal-like patch artifacts [14]. Essentially, these traditional methods can hardly perform well under heavy noise or when images contain complex structures. Moreover, traditional methods can sometimes be computationally expensive [15], which limits their applicability in practical scenarios.

With the recent advancements in computing, deep learning technology has been rapidly developed in various fields [16], [17], [18], including SAR image denoising [19]. Most deep learning-based SAR image denoising methods are performed in a supervised manner [12], [15], [20], [21], [22], [23], [24], [25], [26], [27], which involves designing an appropriate deep convolutional neural network (CNN) architecture and training it using "noisy–clean" image pairs. The performance of denoising SAR images largely depends on the design of the neural network architecture. One of the typical methods is the image-despeckling network (ID-CNN) [20], an eight-layer division residual convolution model. Its goal is to learn the nonlinear mapping relationship between the input noisy SAR image and its containing noise. Finally, the estimated denoised SAR image is produced by performing a division operation between the noisy SAR image and the estimated noise. There are also many more powerful network architectures, e.g., the SAR dilated residual network (SAR-DRN) [21], the spatial and

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transform domain CNN (STD-CNN) [25], the SAR recursive deep CNN prior (SAR-RDCP) [24], the encoder–decoder architecture with a continuous attention module (CAM) [26], and the multiscale residual dense dual attention network (MRDDANet) [12].

Although supervised deep learning-based methods perform well in SAR image denoising, obtaining speckle noise-free SAR images is difficult [24], making it challenging to obtain "noisy–clean" training image pairs using real noisy SAR images. To overcome this challenge, recent methods often rely on simulated data as ground truth to train the network, which consists of the grayscale versions of the plain photograph image dataset (e.g., BSD [25], [28]), optical remote sensing image dataset (e.g., UCML [21], [29]), and the noisy images polluted with artificially added speckle noise [26]. An advantage of using simulated training data is that it can provide mass data to train deep denoising networks. However, due to their distinct imaging mechanisms, the simulated optical noisy and real SAR images differ significantly in terms of image features. For instance, SAR images have unique scattering properties absent in the simulated noisy images. Therefore, denoising networks trained on simulated noisy images exhibit domain gap issues, e.g., unnatural cartoon-like edges, or hallucinations when denoising real SAR images [19]. Furthermore, supervised methods commonly struggle to achieve stable denoising performance across different types of SAR images, including those with varying resolutions, frequencies, and imaging modes. This difficulty arises from the influence of these factors on the disparities between the training and testing data.

In recent years, self-supervised methods have gained attention for their ability to overcome the necessity of clean labels in deep learning-based image denoising. One of the pioneering approaches is Noise2Noise [30], which employs two different noise-contaminated instances of the same clean scene as inputs and labels for training. An extension of Noise2Noise for self-supervised SAR image denoising is the noisy reference-based SAR deep learning (NR-SAR-DL) filter [31], which uses observed noisy SAR images of different phases from the same area. Another approach, Noise2Void [32], uses a particular blind-spot network. The core idea of Noise2Void is to remove the central pixel of the receptive field and allow the network to estimate it by only using its surrounding domain pixels, which serve as the label values. This approach requires only single noisy images for training and eliminates the need for multiple instances of noise pollution. Speckle2Void [33] is a method that applies Noise2Void to the self-supervised denoising of SAR images. Both Noise2Void and Speckle2Void are effective with single noisy images for training. More recent methods, e.g., Neighbor2Neighbor (NB2NB) [13], Blind2Unblind (B2UB) [34], and Self2Self [35], also perform self-supervised denoising under similar data conditions. These methods employ different masking techniques on single noisy images to generate image pairs for training, e.g., neighbor subsampler, global-aware mask mapper, and Bernoulli sampling.

The above self-supervised denoising methods have successfully enabled the training of deep denoising networks for SAR images under realistic conditions, especially when only noisy SAR images without corresponding clean label images are available. Furthermore, these self-supervised methods effectively address the issue of domain gap by ensuring that both training and test data belong to the same data domain, eliminating the need for specific requirements concerning factors such as image resolution, frequency, imaging mode, and system noise. However, these methods typically rely on pixel neighborhood information, which can cause difficulties for CNNs in accurately distinguishing between detailed features and speckle noise. In other words, to achieve adequate noise removal, the denoised images obtained by the aforementioned methods relying on pixel neighborhood information often result in an unavoidable loss of detailed features. This phenomenon occurs because the CNNs adopted by these methods mainly depend on a specific receptive field of the processed image and lack supervision from the clean image pixels. Moreover, the denoised images become blurred when observing visual details, particularly pronounced when denoising SAR images. This is partly because SAR images differ greatly from optical images in terms of visual observation. In addition, speckle noise introduces a higher degree of pollution to SAR images than the noise present in optical images. In general, it is difficult to train an SAR image denoising model with superior detail feature preservation ability by relying solely on the information of the image to be denoised itself, i.e., neighborhood pixel information.

In this work, besides the neighborhood pixel information of the image itself, we introduce semantic information to promote the performance of detail preservation in the self-supervised denoising for SAR images, which refers to the understanding of the observed scene. The semantic information can be obtained through image segmentation, which involves pixel-level semantic label prediction. To this end, we propose a segmentation-guided semantic-aware self-supervised denoising method for SAR images, namely, SARDeSeg, to address the issue of detail feature loss in existing self-supervised denoising methods. Unlike those methods that mainly rely on neighborhood pixel information within the receptive field of CNNs, SARDeSeg incorporates a segmentation network employed to provide semantic guidance during the self-supervised training process of the denoising network. The main contributions of this article are given as follows.

1) We perform self-supervised denoising of SAR images without requiring clean labels. We propose an edge-aware smoothing loss along the horizontal and vertical directions, which is introduced to cooperate with the self-supervised denoising loss to improve speckle noise removal while preserving edge features.
2) We devise a segmentation network as a guide, trained jointly with the self-supervised denoising network. The segmentation network receives denoised images, and the segmentation loss also affects the self-supervised denoising network. Therefore, the denoising network can learn to preserve the detailed features in SAR images.
3) We propose a wavelet transform-based CNN connector that can effectively transmit semantic information from the segmentation network to the self-supervised denoising network by adaptively focusing on different frequency features of SAR images.

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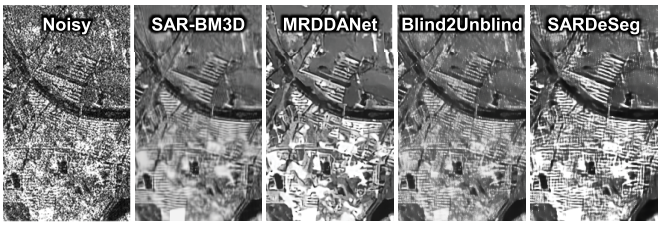


Fig. 1. Examples of noisy SAR images and the corresponding denoised results using SOTA denoising methods, including SAR-BM3D [11], MRDDANet [12], NB2NB [13], and our proposed SARDeSeg.

4) We evaluate the performance of the proposed SARDeSeg through subjective visual observation and objective metric comparison. Extensive experimental results (shown as an example in Fig. 1) demonstrate that SARDeSeg outperforms state-of-the-art (SOTA) methods in SAR image denoising, particularly its ability to preserve detailed edge features.

The remainder of this article is organized as follows. Section II presents our proposed SARDeSeg, including the self-supervised denoising framework for SAR images, the wavelet transform-based connector (WT-Connect), the segmentation guidance (SG), and the edge-aware smoothing loss. Section III reports and discusses the experimental results, along with comparisons with SOTA methods. Finally, Section IV concludes this article.

II. Proposed Method

A. Overview

As an end-to-end framework, the proposed SARDeSeg employs semantic awareness and effective speckle noise suppression for enhancing denoising performance. The overall training procedure is presented in Fig. 2, which consists of two main components.

1) SAR image self-supervised denoising (see details in Section II-B) serves as the fundamental functional part that utilizes only noisy SAR images for basic denoising, which is achieved by modifying the Self2Self model to make it suitable for SAR images and is guided by our proposed self-supervised denoising loss and an edge-aware smoothing loss.
2) Segmentation-guided semantic awareness (see details in Section II-C), aiming to improve the preservation of detailed edge features, is the core enhancement part, which involves a denoising-segmentation connector and a segmentation network, where the connector takes the output of the denoising network and feeds it into the segmentation network. The segmentation loss is calculated between the output of the segmentation network and its corresponding label.

The entire framework is trained end-to-end, with each subnetwork collaborating in the training process. As a result, the segmentation network acts as a guide to enable the denoising network to become semantically aware of SAR images, thereby facilitating effective speckle noise suppression while preserving detailed features.

B. Self-Supervised Denoising Framework

In this work, a modified Self2Self [35] model is employed as the baseline method for self-supervised denoising of SAR images. The self2Self model is selected due to its straightforward effectiveness on self-supervised denoising, using only single noisy SAR images rather than multitemporal data in Noise2Noise [31] as inputs, and without utilizing decorrelate preprocessing in Noise2Void [33]. We modify the original Self2Self model to account for SAR image characteristics, e.g., strong speckle noise and high-frequency feature intensity. The modification is necessary since the Self2Self model was originally developed for self-supervised denoising of optical images, which differ significantly from SAR images. Building on this baseline method, our proposed SARDeSeg aims to enhance the preservation of detailed edge features further, as described in Section II-C.

Concretely, the Self2Self [35] model has demonstrated the ability to effectively train denoising models using pairs of noisy images sampled from a given noisy image as training data. Let $\boldsymbol{y} \in \mathbb{R}^{m\times n}$ and $\boldsymbol{x} \in \mathbb{R}^{m\times n}$ be the noisy and underlying clean SAR images, respectively, where $m \times n$ is the image size. The relationship between $\boldsymbol{y}$ and $\boldsymbol{x}$ is commonly defined as [12], [15], [20], [21], [22], [23], [24], [25], [26]

$$\boldsymbol{y} = \boldsymbol{x} \odot \boldsymbol{n} \tag{1}$$

where $\odot$ denotes the elementwise Hadamard product. $\boldsymbol{n} \in \mathbb{R}^{m\times n}$ is the speckle noise, which follows a gamma distribution with a probability density function (pdf) given by [12], [15], [20], [21], [22], [23], [24], [25], [26]:

$$p(\boldsymbol{n}) = \frac{1}{\Gamma(L)} L^L \boldsymbol{n}^{L-1} e^{-L\boldsymbol{n}}, \quad \boldsymbol{n} \geq 0, \ L \geq 1 \tag{2}$$

where $\Gamma(\cdot)$ denotes the gamma function and L is the number of looks. It is worth noting that the underlying clean SAR image $\boldsymbol{x}$ is unknown and not available in this work, i.e., self-supervised denoising. As shown in Fig. 2, the generated noisy "input-label" image pairs can be obtained through a binary Bernoulli sampling masking, as expressed by

$$(\hat{\boldsymbol{y}}, \tilde{\boldsymbol{y}}) = (\boldsymbol{y} \odot \boldsymbol{b}, \boldsymbol{y} \odot (\boldsymbol{1} - \boldsymbol{b})) \tag{3}$$

where $\boldsymbol{b} \in \{0, 1\}^{m\times n}$ represents the binary Bernoulli instance used for masking, and its settings and pixel selection probabilities adhere to those specified in the original Self2Self [35]. Then, a deep denoising network $\mathcal{F}_{\text{de}}(\cdot)$ with a classic symmetrical encoder–decoder architecture can then be trained, as shown in Fig. 3. The original Self2Self uses max pooling to implement the downsampling operation, also known as feature extraction and denoising. However, the noise pollution in SAR images is usually more substantial than that in optical images. Hence, max pooling is insufficient and may cause speckle noise residues in low-frequency areas of SAR images, e.g., dark homogeneous sea surfaces. For this reason, we replace max pooling with average pooling to improve the smoothing of speckle noise in low-frequency areas of SAR images. To mitigate the issues of gradient disappearance and facilitate faster convergence during network training, layer normalization [36] is added after each convolution operation. In addition, the Self2Self model inserts dropout [37] layers to each convolution

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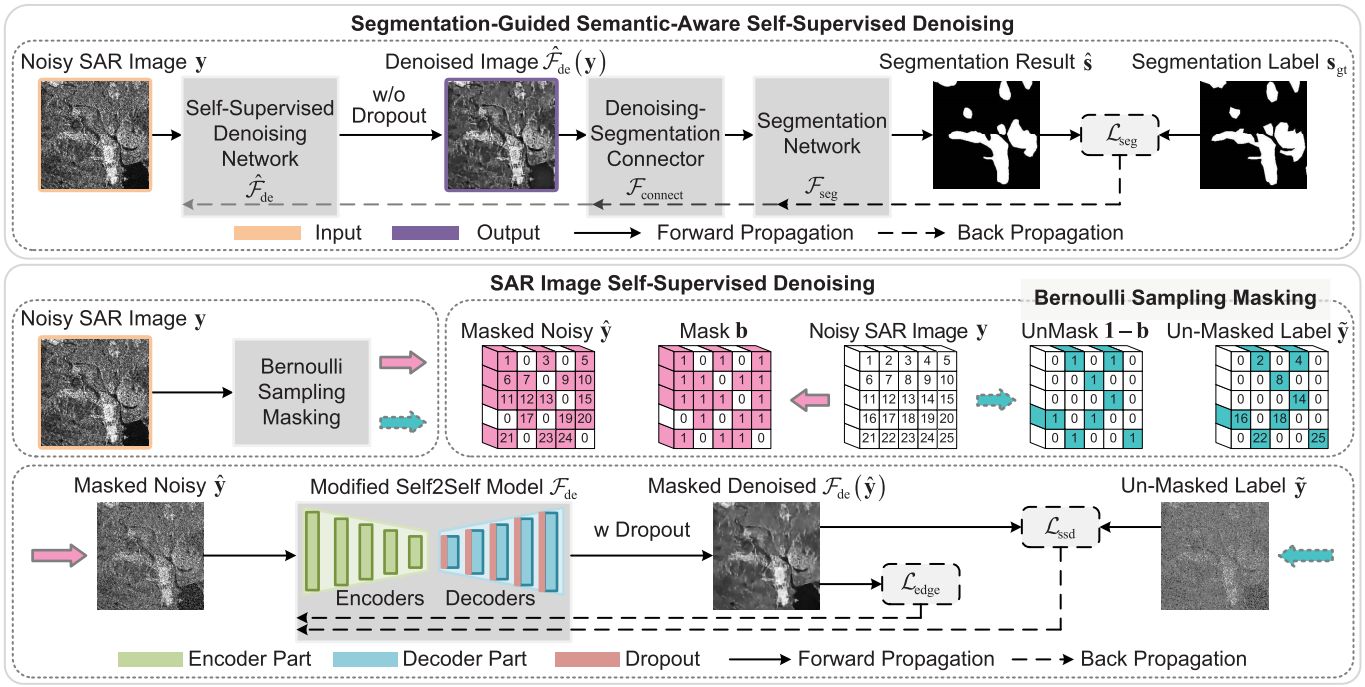


Fig. 2. Illustration of our proposed SARDeSeg. (Top) Segmentation-guided semantic-aware denoising training process. (Bottom) SAR image self-supervised denoising training process. Our proposed SARDeSeg method only requires noisy SAR images for self-supervised denoising training and utilizes the corresponding segmentation labels to provide semantic guidance for denoising.

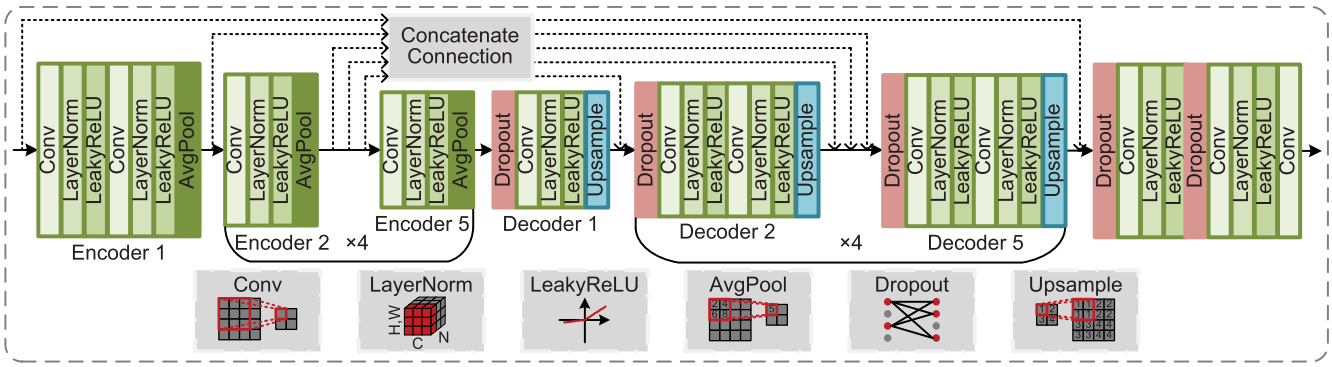


Fig. 3. Network architecture of the modified Self2Self model used for self-supervised denoising of SAR images. Following Self2Self [35] and other self-supervised denoising methods [13], [34], a symmetric architecture, including five encoder–decoder pairs, is adopted. Each encoder employs average pooling to suppress speckle noise. LayerNorm [36] is incorporated after each convolution to accelerate the network training convergence. Dropout [37] is added before each convolution in each decoder to reduce the variance of the denoising prediction.

layer in each decoder to reduce the variance of the denoised prediction.

The objective of the training process is to optimize the parameters θ_{de} of $\mathcal{F}_{\text{de}}(\cdot)$ by minimizing the sum of the loss function between the "output-label" image pair. The original Self2Self utilizes the ℓ_2 norm as the loss function, which can efficiently converge but may result in a significant difference in values affecting the high-frequency feature intensity in SAR images. Specifically, when employing the ℓ_2 norm as the loss function, there is a tendency to prioritize the penalization of predictions with larger bias values. Considering SAR images, the presence of multiplicative speckle noise introduces a substantial bias in pixel points with high-intensity values. Consequently, during the optimization process of the deep denoising network, which aims to minimize the loss function, the intensity values of pixels with high intensity in the predicted denoised image tend to decrease, resulting in a visual effect where strong edge features appear darker. Therefore, we propose a new approach that combines the ℓ_1 norm and ℓ_2 norm as the primary and secondary metrics, respectively, to leverage the advantages of both. The ℓ_1 norm applies the same degree of penalty to all prediction biases to maintain the pixel intensity of strong edge and texture features during SAR image denoising. The ℓ_2 norm benefits the convergence of deep denoising network training. Specifically, the used self-supervised denoising loss $\mathcal{L}_{\text{ssd}}$ can be written as

$$\mathcal{L}_{\text{ssd}}(\theta_{\text{de}}) = \alpha \left\| \left(\mathcal{F}_{\text{de}}(\hat{\boldsymbol{y}}) \odot (\boldsymbol{1}-\boldsymbol{b})\right) - \left(\tilde{\boldsymbol{y}} \odot (\boldsymbol{1}-\boldsymbol{b})\right) \right\|_1 + (1-\alpha) \left\| \left(\mathcal{F}_{\text{de}}(\hat{\boldsymbol{y}}) \odot (\boldsymbol{1}-\boldsymbol{b})\right) - \left(\tilde{\boldsymbol{y}} \odot (\boldsymbol{1}-\boldsymbol{b})\right) \right\|_2 \tag{4}$$

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where $\|\cdot\|_1$ and $\|\cdot\|_2$ denote the ℓ_1 norm and ℓ_2 norm, respectively. $\alpha \in [0, 1]$ balances the contribution of the ℓ_1 norm and ℓ_2 norm. It is worth noting that only the masked pixels are included in $\mathcal{L}_{\text{ssd}}$ measurement. The well-trained denoising network $\mathcal{F}_{\text{de}}(\cdot)$ using $(\hat{\boldsymbol{y}}, \tilde{\boldsymbol{y}})$ is very close to that using the underlying clean labels when enough of such generated $(\hat{\boldsymbol{y}}, \tilde{\boldsymbol{y}})$ can be used.

During inference in the original Self2Self, multiple masked instances of the same noisy image are generated, and each instance is passed through the denoising network with dropout [37] enabled. The corresponding multiple denoised masked outputs are then averaged to produce the final prediction result, which is both computationally complex and time-consuming. In contrast, our proposed approach enables dropout only during training and freezes it during inference. The complete noisy image $\boldsymbol{y}$ is passed through the network only once to obtain the final denoised result $\hat{\boldsymbol{x}}$, which is expressed as

$$\hat{\boldsymbol{x}} = \hat{\mathcal{F}}_{\text{de}}(\boldsymbol{y}) \tag{5}$$

where $\hat{\mathcal{F}}_{\text{de}}(\cdot)$ denotes the dropout-frozen denoising network.

Our proposed SARDeSeg is based on the modified Self2Self model and differs from it by incorporating segmentation-guided semantic awareness. In the following, we will provide a detailed description of SARDeSeg, including the representation of the WT-Connect, the SG, and the edge-aware smoothing loss, aiming to make the denoising network aware of semantic information and be enhanced preserving the detailed features.

C. Segmentation-Guided Semantic Awareness

After applying basic self-supervised denoising to SAR images, we incorporate segmentation-guided semantic awareness. The self-supervised denoising network denoises the image, which is then input to a WT-Connect. The output of this connector is fed to a semantic segmentation network. The segmentation loss is backpropagated, allowing the self-supervised denoising network to become semantically aware and improve its denoising performance. To further improve speckle noise smoothing while preserving edge features, we introduce a new edge-aware smoothing loss.

1) Wavelet Transform-Based Denoising-Segmentation CNN Connector: Image denoising is a low-level computer vision task that maps pixels of input noisy images to pixels of label images, while image semantic segmentation is a high-level task that involves assigning class probabilities to each pixel in an image. In this work, we propose a wavelet transform-based denoising-segmentation CNN connector $\mathcal{F}_{\text{connect}}(\cdot)$ inspired by [38], as shown in Fig. 4.

To enable effective transmission of semantic information from the segmentation network $\mathcal{F}_{\text{seg}}(\cdot)$ to the denoising network $\hat{\mathcal{F}}_{\text{de}}(\cdot)$, we utilize the discrete wavelet transform (DWT) and inverse DWT (IDWT), in combination with convolutional operations. Specifically, as shown in Fig. 5, the output $\hat{\boldsymbol{x}}$ of the denoising network $\hat{\mathcal{F}}_{\text{de}}(\cdot)$ is decomposed using the DWT into a low-frequency approximation coefficient and three high-frequency (i.e., vertical, horizontal, and diagonal)

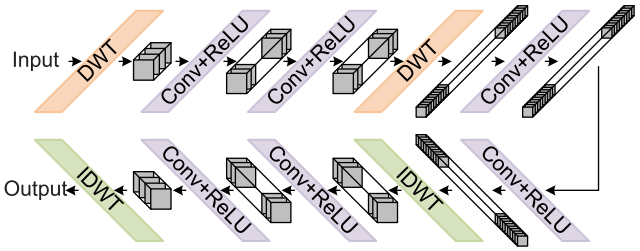


Fig. 4. Network architecture of the wavelet transform-based denoising-segmentation CNN connector.

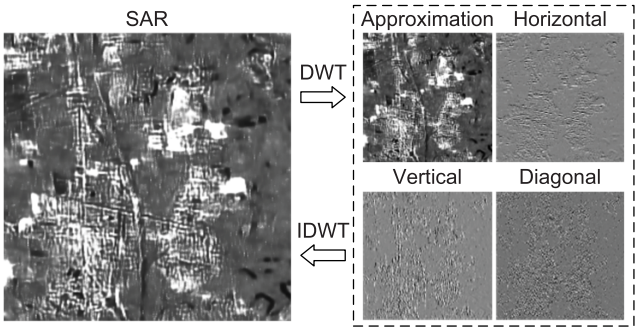


Fig. 5. Illustration of the (inverse) discrete wavelet transform process for converting an SAR image into its approximation, horizontal, vertical, and diagonal detail coefficients, and vice versa.

coefficients, denoted as $\hat{\boldsymbol{x}}_{(k)}$, where $k \in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\}$. Each coefficient is then further decomposed using the DWT into four subcoefficients, i.e., $\{\hat{\boldsymbol{x}}_{(k,\text{LL})}, \hat{\boldsymbol{x}}_{(k,\text{LH})}, \hat{\boldsymbol{x}}_{(k,\text{HL})}, \hat{\boldsymbol{x}}_{(k,\text{HH})}\}$. These subcoefficients are subsequently reconstructed into the coefficients $\boldsymbol{x'}_{(k)}$ and then reconstructed into the image $\boldsymbol{x'}$ using two IDWTs.

To facilitate feature extraction, we incorporate convolutions with rectified linear unit (ReLU) activations in adjacent transform operations. The two-level coefficient decomposition process of the connector can be mathematically expressed as

$$\begin{aligned} \left\{\hat{\boldsymbol{x}}_{(\text{LL})}, \hat{\boldsymbol{x}}_{(\text{LH})}, \hat{\boldsymbol{x}}_{(\text{HL})}, \hat{\boldsymbol{x}}_{(\text{HH})}\right\} &= \varphi_1\left(\hat{\boldsymbol{x}}\right) \\ \left\{\hat{\boldsymbol{x}}_{(k,\text{LL})}, \hat{\boldsymbol{x}}_{(k,\text{LH})}, \hat{\boldsymbol{x}}_{(k,\text{HL})}, \hat{\boldsymbol{x}}_{(k,\text{HH})}\right\} &= \varphi_2\left(f_{\text{cr}}\left(f_{\text{cr}}\left(\hat{\boldsymbol{x}}_{(k)}\right)\right)\right) \\ & \qquad k \in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\} \end{aligned} \tag{6}$$

where $\varphi_1(\cdot)$ and $\varphi_2(\cdot)$ denote the first- and second-level DWT operations, respectively, and $f_{\text{cr}}(\cdot)$ represents the feature facilitation operation combined the convolution and ReLU. Then, the two-level coefficient reconstruction process can be mathematically expressed as

$$\begin{aligned} \boldsymbol{x'}_{(k)} &= \psi_2\left(f_{\text{cr}}\left(f_{\text{cr}}\left(\left\{\hat{\boldsymbol{x}}_{(k,\text{LL})}, \hat{\boldsymbol{x}}_{(k,\text{LH})}, \hat{\boldsymbol{x}}_{(k,\text{HL})}, \hat{\boldsymbol{x}}_{(k,\text{HH})}\right\}\right)\right)\right) \\ \boldsymbol{x'} &= \psi_1\left(f_{\text{cr}}\left(f_{\text{cr}}\left(\left\{\boldsymbol{x'}_{(\text{LL})}, \boldsymbol{x'}_{(\text{LH})}, \boldsymbol{x'}_{(\text{HL})}, \boldsymbol{x'}_{(\text{HH})}\right\}\right)\right)\right) \\ & \qquad k \in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\} \end{aligned} \tag{7}$$

where $\psi_1(\cdot)$ and $\psi_2(\cdot)$ denote the first- and second-level IDWT operations, respectively. The two-level DWT-IDWT approach can achieve a more detailed coefficient decomposition of different frequencies. Therefore, different degrees of consideration can be adaptively performed on coefficients of different frequencies, allowing the semantic information provided by

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the segmentation network to be effectively transmitted to the denoising network.

2) Segmentation Guidance: In the end-to-end training process of the whole framework, the input noisy SAR images are processed through the self-supervised denoising network and the denoising-segmentation connector. The processed image is then fed into a semantic segmentation network $\mathcal{F}_{\text{seg}}(\cdot)$, which aims to learn and provide semantic understanding of the information about the input SAR image through pixel-level prediction. The semantic segmentation network assigns a semantic label to each image pixel, resulting in a semantic segmentation result $\hat{\boldsymbol{s}}$. To achieve this, the popular SegFormer [39] model is adopted as the semantic segmentation network. The process can be expressed as

$$\hat{\boldsymbol{s}} = \mathcal{F}_{\text{seg}}\left(\mathcal{F}_{\text{connect}}\left(\hat{\mathcal{F}}_{\text{de}}(\boldsymbol{y})\right)\right). \tag{8}$$

The source of the semantic information throughout the framework is obtained using the segmentation loss function $\mathcal{L}_{\text{seg}}$, which calculates the cross entropy between the segmentation result $\hat{\boldsymbol{s}}$ and its corresponding segmentation label $\boldsymbol{s}_{\text{gt}} \in \{0, 1\}^{m \times n}$. The segmentation loss can be expressed as

$$\mathcal{L}_{\text{seg}}\left(\theta_{\text{de}}, \theta_{\text{connect}}, \theta_{\text{seg}}\right) = \text{CrossEntropy}\left(\hat{\boldsymbol{s}}, \boldsymbol{s}_{\text{gt}}\right) \tag{9}$$

where θ_{connect} and θ_{seg} are the optimized parameters of $\mathcal{F}_{\text{connect}}(\cdot)$ and $\mathcal{F}_{\text{seg}}(\cdot)$, respectively.

3) Edge-Aware Smoothing Loss Along the Horizontal and Vertical Directions: The total variance (TV) loss is a commonly used regularization term in deep learning-based SAR image denoising methods [40]. It enhances the smoothness of the denoised image by reducing the absolute difference between neighboring pixels, which can be expressed as

$$\mathcal{L}_{\text{tv}}(\theta_{\text{de}}) = \left\|\nabla \mathcal{F}_{\text{de}}(\boldsymbol{y})\right\|_1 \tag{10}$$

where ∇ represents the gradient operator. However, TV loss can result in the blurring of sharply contrasting edges. To address this, we propose an edge-aware smoothing loss along the horizontal and vertical directions, formalized as

$$\begin{aligned}\mathcal{L}_{\text{edge}}(\theta_{\text{de}}) = &\left\|\nabla_{\text{h}} \mathcal{F}_{\text{de}}\left(\hat{\boldsymbol{y}}\right)\right\|_1 \cdot e^{-\left\|\nabla_{\text{h}} \mathcal{F}_{\text{de}}\left(\hat{\boldsymbol{y}}\right)\right\|_1} \\ &+ \left\|\nabla_{\text{v}} \mathcal{F}_{\text{de}}\left(\hat{\boldsymbol{y}}\right)\right\|_1 \cdot e^{-\left\|\nabla_{\text{v}} \mathcal{F}_{\text{de}}\left(\hat{\boldsymbol{y}}\right)\right\|_1}\end{aligned} \tag{11}$$

where ∇_{h} and ∇_{v} represent gradient operators along the horizontal and vertical directions, respectively. In this case, the absolute neighborhood difference is scaled by the exponential of the negative difference. Furthermore, the horizontal and vertical smoothing terms are considered separately to refine optimization. Fig. 6 shows the difference between TV- and edge-aware smoothing losses. In the initial stage of increasing absolute neighborhood difference, corresponding to the enhancement of speckle noise in homogeneous areas, both losses exhibit a rising trend. The ratio of the TV loss value to the difference is always one, while the ratio of the edge-aware loss decreases as the difference increases. In the latter stage, which pertains to the presence of sharp edges in the area, the TV loss value increases significantly, leading to edge blurring. In contrast, the edge-aware loss value tends to be stable, thus preserving edge sharpness.

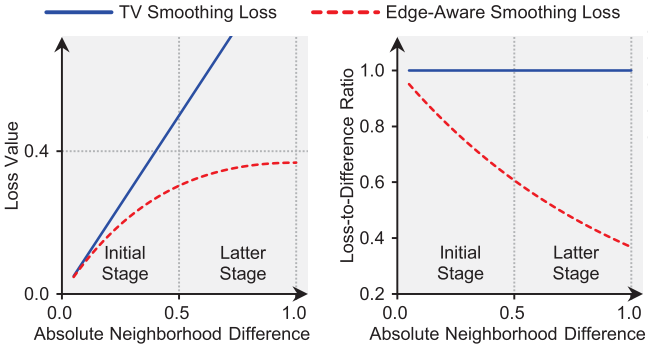


Fig. 6. Comparison of TV and edge-aware smoothing losses, in terms of (left) loss values and (right) loss-to-difference ratios, relative to the absolute neighborhood difference.

4) Total Loss Function: So far, the components of our proposed SARDeSeg method have been introduced, including three subnetworks: a self-supervised denoising network $\mathcal{F}_{\text{de}}(\cdot)$, a wavelet transform-based denoising-segmentation connector $\mathcal{F}_{\text{connect}}(\cdot)$, and a segmentation network $\mathcal{F}_{\text{seg}}(\cdot)$. These subnetworks are trained and optimized jointly in an end-to-end manner. The total loss function consists of three components: the self-supervised denoising loss $\mathcal{L}_{\text{ssd}}$, the edge-aware smoothing loss $\mathcal{L}_{\text{edge}}$, and the segmentation loss $\mathcal{L}_{\text{seg}}$, which can be expressed as

$$\begin{aligned}\mathcal{L}_{\text{total}}\left(\theta_{\text{de}}, \theta_{\text{connect}}, \theta_{\text{seg}}\right) = &\mathcal{L}_{\text{ssd}}(\theta_{\text{de}}) + \lambda_{\text{edge}} \mathcal{L}_{\text{edge}}(\theta_{\text{de}}) \\ &+ \lambda_{\text{seg}} \mathcal{L}_{\text{seg}}\left(\theta_{\text{de}}, \theta_{\text{connect}}, \theta_{\text{seg}}\right)\end{aligned} \tag{12}$$

where λ_{edge} and λ_{seg} are the weighting coefficients of $\mathcal{L}_{\text{edge}}$ and $\mathcal{L}_{\text{seg}}$, respectively.

The main implementation steps are summarized in Algorithm 1. Among them, self-supervised denoising loss $\mathcal{L}_{\text{ssd}}$ and edge-aware smoothing loss $\mathcal{L}_{\text{edge}}$ guide the training of the self-supervised denoising network and update its parameters θ_{de}. In contrast, the segmentation loss guides the optimization and updates the parameters θ_{de}, θ_{connect}, and θ_{seg} of all three subnetworks. During the training process, the segmentation loss is sequentially backpropagated through the segmentation network, connector, and denoising network, to ensure the successful transmission of semantic information to the denoising network. As a result, the denoising network can achieve speckle noise suppression while preserving more details of SAR images due to its semantic awareness.

III. Experimental Results and Discussion

In this section, we first provide details of the experimental setup, including implementation details and evaluation metrics. We then perform a comparative analysis of SARDeSeg with SOTA SAR image denoising methods. To further validate the effectiveness of SARDeSeg, we conduct ablation studies on several critical design choices. Finally, we make an extended attempt to transfer our proposed SARDeSeg to other SAR sensor images and provide a runtime comparison.

A. Experimental Setup

1) Dataset: To verify the effectiveness of the proposed SARDeSeg, we conduct denoising experiments on images with

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Algorithm 1 SARDeSeg

Training

Input: Noisy SAR images $\{\boldsymbol{y}^{(n)}, n = 1, 2, \cdots, N\}$ and the corresponding segmentation labels $\{\boldsymbol{s}_{\text{gt}}{}^{(n)}, n = 1, 2, \cdots, N\}$.

Output: Dropout-frozen denoising network $\hat{\mathcal{F}}_{\text{de}}(\cdot)$ with well-trained parameters θ_{de}.

Initialize the denoising network $\mathcal{F}_{\text{de}}(\cdot)$, connector network $\mathcal{F}_{\text{connect}}(\cdot)$, and segmentation network $\mathcal{F}_{\text{seg}}(\cdot)$ with the learnable parameters θ_{de}, θ_{connect}, and θ_{seg};
while *epoch_num* < *max_epoch_num* **do**
  for $n = 1$ to N **do**
    Generate the random binary Bernoulli instance $\boldsymbol{b}$;
    $(\hat{\boldsymbol{y}}^{(n)}, \tilde{\boldsymbol{y}}^{(n)}) \leftarrow (\boldsymbol{y}^{(n)} \odot \boldsymbol{b}, \boldsymbol{y}^{(n)} \odot (\boldsymbol{1} - \boldsymbol{b}))$;
    $\mathcal{L}_{\text{ssd}} \leftarrow$ (4) $\left\langle \mathcal{F}_{\text{de}}(\hat{\boldsymbol{y}}^{(n)}), \tilde{\boldsymbol{y}}^{(n)} \right\rangle$;
    $\mathcal{L}_{\text{edge}} \leftarrow$ (11) $\left\langle \mathcal{F}_{\text{de}}(\hat{\boldsymbol{y}}^{(n)}) \right\rangle$;
    $\hat{\mathcal{F}}_{\text{de}}(\cdot) \leftarrow$ freeze the dropout in $\mathcal{F}_{\text{de}}(\cdot)$;
    $\hat{\boldsymbol{s}}^{(n)} \leftarrow \mathcal{F}_{\text{seg}}(\mathcal{F}_{\text{connect}}(\hat{\mathcal{F}}_{\text{de}}(\boldsymbol{y}^{(n)})))$;
    $\mathcal{L}_{\text{seg}} \leftarrow \text{CrossEntropy}(\hat{\boldsymbol{s}}^{(n)}, \boldsymbol{s}_{\text{gt}}{}^{(n)})$;
    Backward $\mathcal{L}_{\text{ssd}}$ and $\mathcal{L}_{\text{edge}}$ to update θ_{de};
    Backward $\mathcal{L}_{\text{seg}}$ to update θ_{de}, θ_{connect}, and θ_{seg}.
  end for
end while

Test

Input: Noisy SAR image $\boldsymbol{y}$ and dropout-frozen denoising network $\hat{\mathcal{F}}_{\text{de}}(\cdot)$ with well-trained parameters θ_{de}.

Output: Denoised SAR image $\hat{\boldsymbol{x}}$.

$\hat{\boldsymbol{x}} \leftarrow \hat{\mathcal{F}}_{\text{de}}(\boldsymbol{y})$.

synthetic speckle noise. The noisy images with single-look speckle noise are constructed using the BSD dataset [28] according to (1) and (2) for both training and evaluating the denoising network. Then, we conduct comparative and ablation experiments on the Gaofen-3 satellite images from the publicly available SARBuD dataset [41], which contains 25 000 SAR images along with the corresponding segmentation labels provided by expert interpreters.[1] Each sample image size is 256 × 256 pixels. The dataset is randomly partitioned into training, validation, and testing sets, consisting of 15 000, 5000, and 5000 samples, respectively. During the training phase, we apply data augmentation techniques, including horizontal and vertical flipping, geometric rotations (0°, 90°, 180°, and 270°) for the self-supervised denoising stage, and random rotations (arbitrary rotations) and random scale cropping (with a scale range of [0.375, 1.0] and a crop resolution of 128 × 128) for the segmentation stage. In addition, to illustrate the applicability of the proposed SARDeSeg, we employ TerraSAR-X[2] satellite images to conduct SAR transfer denoising experiments, where the denoising model is trained on Gaofen-3 images and directly applied to denoise TerraSAR-X images.

2) Implementation Details: We optimize the proposed SARDeSeg using the adaptive moment estimation (ADAM) [42] and train it for 50 epochs with a batch size of 32. We employ the POLY learning rate adjustment strategy,[3] with an initial learning rate of 10^{-3} and a final learning rate of 10^{-5}. The network architecture of $\mathcal{F}_{\text{seg}}(\cdot)$ is the SegFormer with the MiT-b5[4] as the encoder. The pixel selection probability of mask $\boldsymbol{b}$ in (3) follows the original Self2Self [35] with a setting of 0.3. The weighting coefficients (α, λ_{edge}, and λ_{seg}) of the ℓ_1 norm, edge-aware smoothing loss, and segmentation loss in (4) and (12) are set to 0.9, 10^{-6}, and 5×10^{-3}, respectively. We implement SARDeSeg using the PyTorch framework [43] and train it on a computer with a GeForce GTX 3090 GPU.

3) Evaluation Metrics: We evaluate the performance of our proposed SARDeSeg method using several widely adopted metrics in the field of SAR image denoising, i.e., peak signal-to-noise ratio (PSNR), equivalent number of looks (ENL) [11], [26], figure of merit (FOM) [25], [44], ratio gradient preservation index (RGPI) [45], and mean of ratio (MoR) [26], [46]. To evaluate the denoising performance on images with synthetic speckle noise, we employ PSNR to quantify the differences between the denoised result and the corresponding clean label. A higher PSNR value indicates a more effective recovery of clean images. For the evaluation on real SAR images, ENL and MoR within the homogeneous areas of the denoised results are calculated, which are defined as

$$\text{ENL} = \text{mean}\left(\hat{\boldsymbol{x}}_{\text{homo}}\right)^2 \Big/ \text{std}\left(\hat{\boldsymbol{x}}_{\text{homo}}\right)^2 \tag{13}$$

$$\text{MoR} = \text{mean}\left(\boldsymbol{y}_{\text{homo}} \Big/ \hat{\boldsymbol{x}}_{\text{homo}}\right) \tag{14}$$

where $\boldsymbol{y}_{\text{homo}}$ and $\hat{\boldsymbol{x}}_{\text{homo}}$ represent the homogeneous areas of the noisy and denoised images, respectively, and mean(·) and std(·) denote the mean and standard deviation calculations, respectively. In particular, a higher ENL value indicates more effective smoothing of speckle noise. In addition, an MoR value closer to 1 indicates better radiometric preservation performance. Regarding FOM and RGPI, FOM is calculated within the areas containing edge features, while RGPI is calculated throughout the entire image. The detailed definitions of FOM and RGPI can be found in [44] and [45], respectively. An FOM value closer to 1 suggests more complete preservation of image details in the denoised image, while a higher RGPI value implies improved preservation of edge features.

4) Comparison Methods: We compare the performance of our proposed SARDeSeg with several SOTA SAR image denoising methods, including traditional probabilistic patch-based (PPB) filter [10], SAR-block-matching 3-D (SAR-BM3D) algorithm [11], supervised deep learning-based SAR-DRN [21], SAR-RDCP [24], multiobjective network (MONet) [27], and MRDDANet [12], as well as the latest self-supervised deep learning-based NB2NB [13] and B2UB [34]. In the literature, PPB and SAR-BM3D are often used as benchmarks to assess achievable performance. The supervised methods SAR-DRN, SAR-RDCP, MONet, and MRDDANet require clean labels during training and thus use simulated noisy images, as described in [24]. In the

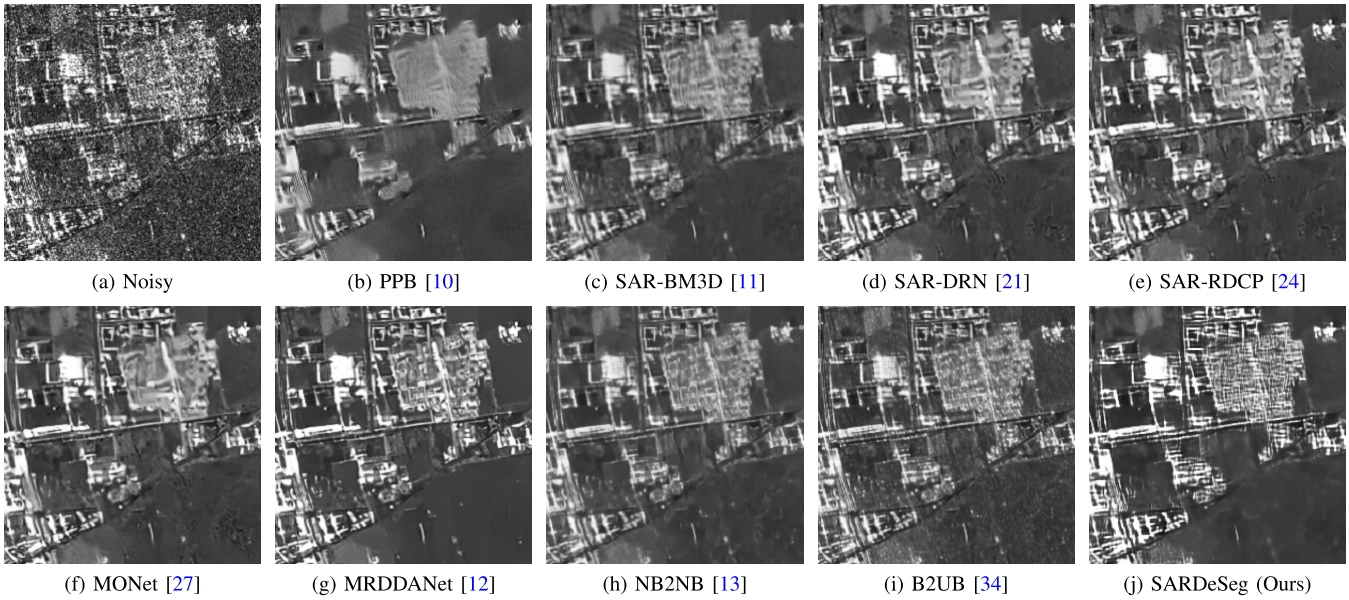
[1] http://github.com/CAESAR-Radi/SARBuD
[2] https://tpm-ds.eo.esa.int/oads/access/collection/TerraSAR-X/tree
[3] http://github.com/cmpark0126/pytorch-polynomial-lr-decay
[4] http://huggingface.co/docs/transformers/model_doc/segformer

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TABLE I

PSNR COMPARISON OF DIFFERENT SAR IMAGE DENOISING METHODS ON IMAGES WITH SYNTHETIC SPECKLE NOISE. BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Image	Noisy	PPB [10]	SAR-BM3D [11]	SAR-DRN [21]	SAR-RDCP [24]	MONet [27]	MRDDANet [12]	NB2NB [13]	B2UB [34]	SARDeSeg
Coastal	9.82	16.19	16.68	21.38	21.38	21.29	**21.52**	21.45	21.47	21.35
Crowd	8.40	13.42	13.85	20.00	20.02	19.68	**20.39**	20.17	20.20	20.14
Dog	10.51	18.53	19.05	22.72	22.70	22.61	22.79	22.61	22.72	**22.80**
Kangaroo	11.68	22.67	23.75	25.35	25.43	25.17	**25.51**	24.91	24.92	25.47
People	10.54	19.61	20.17	23.84	23.87	23.72	**24.05**	23.80	23.87	24.00
Riverbank	9.99	15.58	15.97	18.08	18.07	17.97	**18.19**	18.03	18.07	**18.19**
Tiger	10.49	16.36	16.96	19.31	19.29	19.19	**19.37**	19.12	19.21	19.34
Underwater	11.16	19.83	20.45	22.41	22.39	22.32	**22.44**	22.29	22.36	22.36
Venue	9.22	14.65	14.99	19.98	19.71	19.72	**20.37**	19.40	19.83	19.57
Zebra	9.76	16.96	17.55	20.45	20.16	20.55	20.52	20.24	20.52	**20.88**
Average	10.16	17.38	17.94	21.35	21.30	21.22	**21.51**	21.20	21.32	21.41



(a) Noisy (b) PPB [10] (c) SAR-BM3D [11] (d) SAR-DRN [21] (e) SAR-RDCP [24]

(f) MONet [27] (g) MRDDANet [12] (h) NB2NB [13] (i) B2UB [34] (j) SARDeSeg (Ours)

Fig. 7. Visual comparison of denoising results for the Heilongjiang-1 region image using different methods.

evaluation of denoising real SAR images, the self-supervised methods NB2NB and B2UB share the same training dataset of real SAR images as described in Section III-A1 with our proposed SARDeSeg. The parameters of all compared methods are set as recommended in their respective original articles.

B. Evaluation on Images With Synthetic Speckle Noise

We first compare the denoising performance of our proposed SARDeSeg with other SOTA methods on images with synthetic speckle noise. The PSNR results calculated between the denoising results obtained by each method and the corresponding clean label images are given in Table I. Although the traditional methods (i.e., PPB and SAR-BM3D) exhibit a significant improvement in PSNR values compared to the original noisy images, their performance remains inferior to deep learning-based methods. Among the supervised deep learning-based methods, MRDDANet demonstrates the highest average PSNR value due to its utilization of a deep and complex network architecture. In contrast to two other self-supervised deep learning-based methods (i.e., NB2NB and B2UB), our proposed SARDeSeg achieves a higher average PSNR value. Notably, despite not employing clean label images for supervision during the training process, our proposed SARDeSeg still exhibits comparable PSNR values with supervised deep learning-based methods. Specifically, our proposed SARDeSeg yields a slightly lower average PSNR value of 0.1 compared to MRDDANet and a slightly higher value of 0.06 compared to SAR-DRN, offering a more promising solution where only noisy images are available for training.

C. Evaluation on Real SAR Images

We present a comprehensive performance comparison of real SAR image denoising between our proposed SARDeSeg and other SOTA methods, which includes: 1) overall visual comparison; 2) detailed visual comparison; 3) ENL map estimation comparison; and 4) quantitative comparison.

1) Overall Visual Comparison: The results of our proposed SARDeSeg method are compared with several SOTA SAR image denoising methods in terms of visual quality, as shown

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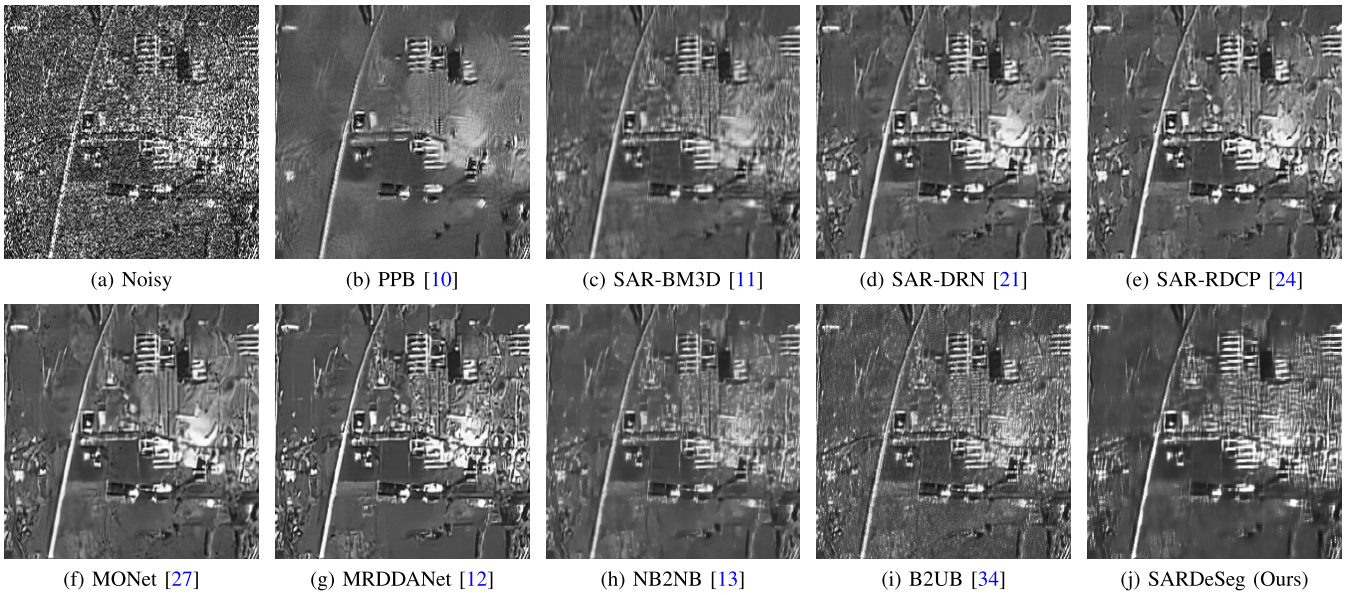


Fig. 8. Visual comparison of denoising results for the Jingjinji-1 region image using different methods.

in Figs. 7 and 8. The traditional PPB and SAR-BM3D methods are able to reduce speckle noise effectively but suffer from oversmoothing, causing a loss of detailed textures and blurring edge features. Supervised deep learning methods (i.e., SAR-DRN, SAR-RDCP, MONet, and MRDDANet) produce noticeably better outcomes than traditional PPB and SAR-BM3D. MRDDANet outperforms the other supervised deep learning techniques with more extensive speckle noise reduction and contrasted edge features due to its more expressive, deeper network architecture. However, these supervised methods often produce cartoon-like artifacts in the denoised results, which should not be present in real SAR images. The artifacts occur when supervised methods use noisy training data simulated from optical images, which significantly differ in image features from real SAR images. The self-supervised methods, i.e., NB2NB and B2UB, produce denoising results that avoid the issue of cartoon-like artifacts and maintain a visual style consistent with the original SAR images. NB2NB exhibits greater speckle noise smoothing than B2UB but sacrifices feature preservation. In comparison, B2UB retains more detailed textures than NB2NB but has noticeable residual noise. It is a challenge for these two methods to simultaneously account for significant noise smoothing and more pronounced retention of detail features, as they have difficulty separating complicated textures from heavy speckle noise by relying solely on information from neighboring pixels. The denoising results obtained from our proposed SARDeSeg avoid the cartoon-like artifact problem observed in supervised deep learning methods (i.e., SAR-DRN, SAR-RDCP, MONet, and MRDDANet) while effectively suppressing most of the speckle noise. In addition, our proposed SARDeSeg also produces images with sharper edges and finely detailed textures, particularly in regions with dense scattering phenomena, which sets it apart from other methods.

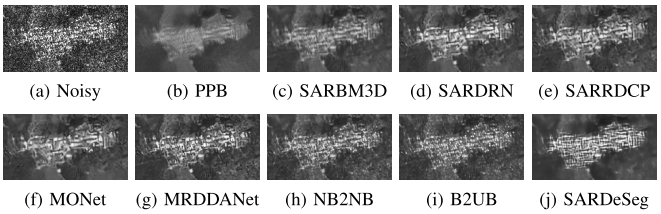


Fig. 9. Visual comparison of denoising results for a local area of the Heilongjiang-2 region image using different methods. This comparison aims to evaluate the preservation ability of texture features.

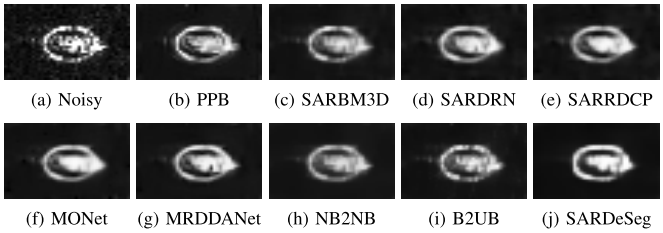


Fig. 10. Visual comparison of denoising results for a local area of the Jingjinji-2 region image using different methods. This comparison aims to evaluate the preservation ability of strong edge features.

2) Detailed Visual Comparison: Figs. 9 and 10 presents visual comparisons of the magnified areas of the denoising results from various methods, to evaluate their ability to preserve texture features and strong edge features in detail. As shown in Fig. 9, PPB loses almost all texture features, while SAR-BM3D performs slightly better. Supervised deep learning methods (i.e., SAR-DRN, SAR-RDCP, MONet, and MRDDANet) outperform traditional methods. However, they convert some finer textures into thicker edge features due to the use of noisy training data simulated by optical images lacking the unique texture features of SAR images. Self-supervised methods (i.e., NB2NB and B2UB) demonstrate

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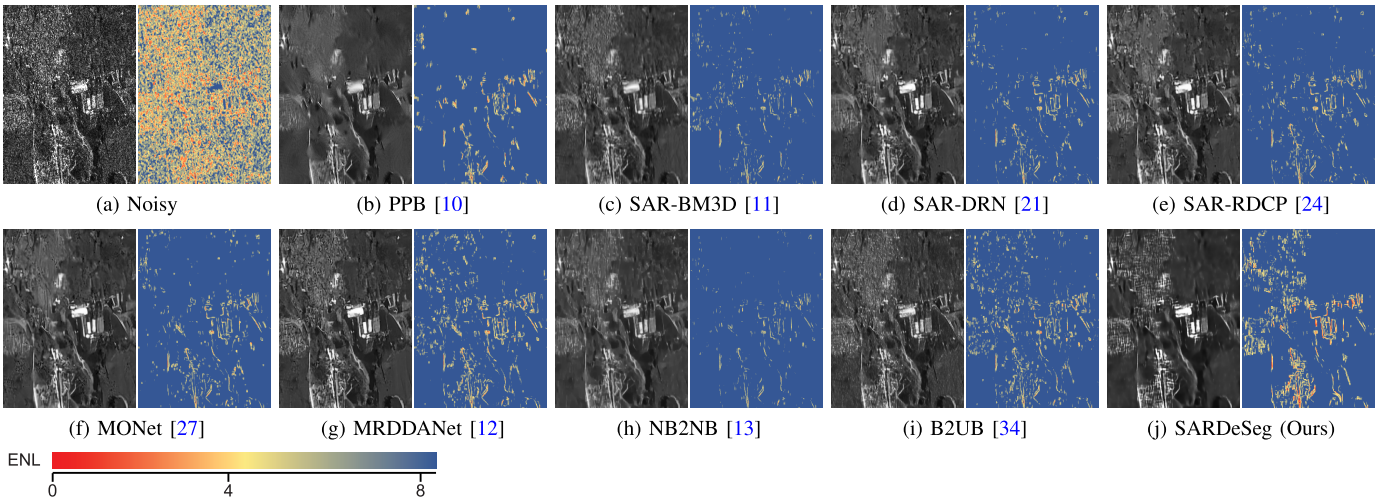


Fig. 11. Visual comparison of denoising results and corresponding ENL maps for the Jingjinji-3 region image using different methods.

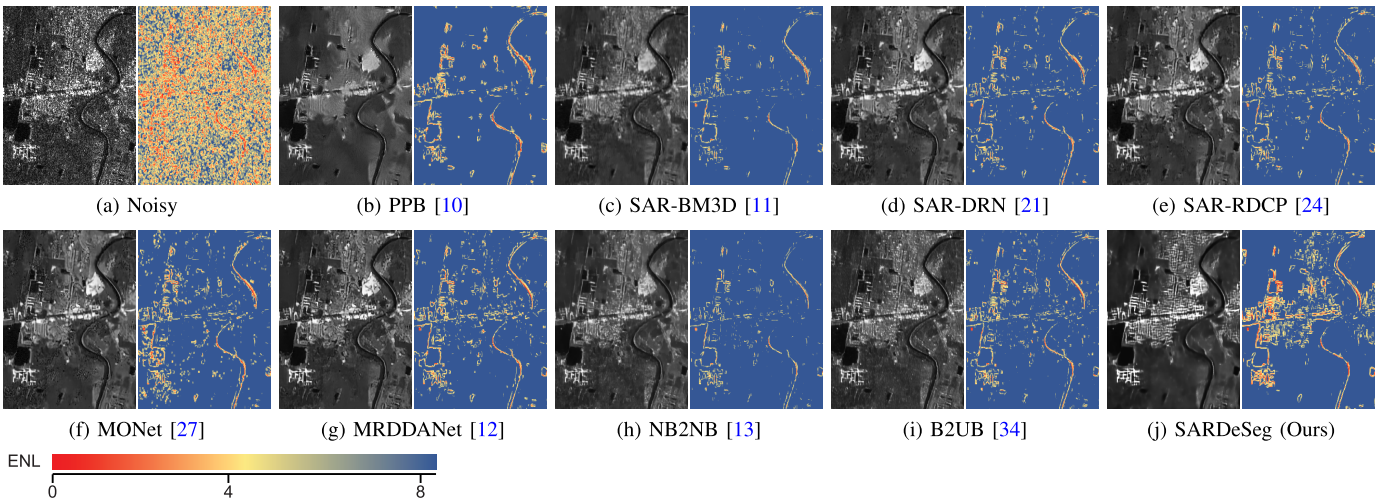


Fig. 12. Visual comparison of denoising results and corresponding ENL maps for the Jingjinji-4 region image using different methods.

better retention of texture features closer to those found in real SAR images. However, noticeable speckle noise still remains in their detailed texture regions. In contrast, the results from our proposed SARDeSeg show the clear preservation of detailed texture features with significant speckle noise reduction, resulting in improved visual clarity. As shown in Fig. 10, both SAR-BM3D and NB2NB tend to reduce image contrast, causing strong edge features to appear darker compared to the original SAR images. In comparison, our proposed SARDeSeg maintains the sharpness of edge features, making them more prominent and easier to observe.

3) ENL Map Estimation Comparison: We introduce an estimation technique called ENL map [24], which allows for explicit observation of speckle noise smoothing and feature preservation in denoised images. The ENL is a measure of the statistical quality of the image, defined as the ratio of the mean intensity to the standard deviation. To generate the ENL map, a sliding 3×3 window is moved across the entire image, and the ENL value is calculated for the area covered by the window at each step. After the window covers the entire image, an ENL map is created, where each pixel in the map represents the ENL value of the corresponding area in the image. In this way, the degree of preservation of edge and texture features in heterogeneous areas and the smoothing of speckle noise in homogeneous areas can be visualized and compared. Figs. 11 and 12 give two sets of comparative illustrations, where the denoised images obtained by different methods and the corresponding ENL maps are displayed. Among the compared methods, in Fig. 11, PPB and NB2NB produce relatively few edge features in their ENL maps, whereas MRDDANet and B2UB perform slightly better. In Fig. 12, PPB, SAR-BM3D, and NB2NB perform poorly, while MONet, MRDDANet, and B2UB show better performance. Our proposed SARDeSeg is found to be comparable to other methods in reducing the speckle noise in homogeneous areas while also exhibiting improved preservation of edge features and texture details in heterogeneous regions, as demonstrated in the ENL maps. This comparison further demonstrates that our proposed SARDeSeg has practical semantic perception capability in SAR image denoising,

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TABLE II

QUANTITATIVE COMPARISON OF DIFFERENT SAR IMAGE DENOISING METHODS. UNDERLINE INDICATES THE BETTER ONE OF THE TRADITIONAL METHODS. BOLD INDICATES THE BETTER ONE OF THE DEEP LEARNING-BASED METHODS

Metric	Data	PPB [10]	SAR-BM3D [11]	SAR-DRN [21]	SAR-RDCP [24]	MONet [27]	MRDDANet [12]	NB2NB [13]	B2UB [34]	SARDeSeg
ENL [11]	*#1*	387.7850	194.4789	68.6832	65.0031	69.1009	72.7855	84.7508	35.2822	**103.5308**
	#2	176.6052	149.8853	68.3310	79.8530	60.6976	105.7032	110.9594	35.6628	**145.0369**
	#3	368.8471	142.8771	56.2535	61.9113	55.3183	130.9118	108.9070	33.3936	**140.4287**
	#4	224.9012	204.3291	71.5633	79.5452	28.4861	90.8284	122.5966	41.9763	**157.0496**
	#5	393.1856	351.5879	84.2454	84.3868	91.4853	153.7903	93.6219	34.6555	**292.5242**
	Avg.	310.2648	208.6316	69.8153	74.1399	61.0177	110.8038	104.1671	36.1941	**167.7140**
FOM [25]	*#1*	0.9361	0.9493	0.8980	0.9305	0.9111	0.9452	0.9550	**0.9590**	0.9531
	#2	0.9286	0.9477	0.9381	0.9457	0.9189	0.9398	0.9402	0.9472	**0.9496**
	#3	0.9015	0.9461	0.9475	0.9555	0.9525	0.9529	0.9479	0.9495	**0.9568**
	#4	0.9269	0.9438	0.9520	0.9523	0.9553	0.9487	0.9559	0.9367	**0.9603**
	#5	0.8858	0.9403	0.9444	0.9384	0.9406	0.9402	0.9546	0.9259	**0.9578**
	Avg.	0.9158	0.9454	0.9360	0.9445	0.9357	0.9454	0.9507	0.9436	**0.9555**
RGPI [45]	*#1*	-0.2633	0.0402	0.0858	0.0981	0.1000	**0.1210**	0.0462	0.1104	0.1034
	#2	-0.2089	0.0431	0.0836	0.0973	0.0942	0.0819	0.0626	0.1160	**0.1227**
	#3	-0.2755	0.0160	0.0812	0.1015	0.0911	0.0728	0.0572	**0.1113**	0.1080
	#4	-0.2286	0.0171	0.0771	0.0869	0.0822	0.0894	0.0505	**0.1022**	0.0975
	#5	-0.2397	0.0155	0.0932	**0.1046**	0.0951	0.0982	0.0689	0.1018	0.0931
	Avg.	-0.2432	0.0264	0.0842	0.0977	0.0925	0.0927	0.0571	**0.1083**	0.1049
MoR [46]	*#1*	0.9583	0.9615	0.9019	0.8991	0.9134	0.9110	0.9648	0.9713	**0.9736**
	#2	0.9568	0.9607	0.8933	0.8848	0.9066	0.9051	0.9655	0.9706	**0.9714**
	#3	0.9701	0.9783	0.9497	0.9485	0.9460	0.9665	0.9835	0.9888	**0.9889**
	#4	0.9630	0.9680	0.9112	0.9116	0.9193	0.9246	0.9700	**0.9743**	0.9710
	#5	0.9583	0.9655	0.8980	0.8903	0.9135	0.9116	0.9658	0.9728	**0.9732**
	Avg.	0.9613	0.9668	0.9108	0.9068	0.9198	0.9237	0.9699	**0.9756**	**0.9756**

enabling the identification and preservation of meaningful SAR image features even in the presence of heavy speckle noise.

4) Quantitative Comparison: Table II presents a comparative analysis of the quantitative metrics described in Section III-A3, computed on several test cases. We conduct a detailed analysis on several levels. First, we examine ENL, where the traditional PPB and SAR-BM3D show significantly higher values than those of deep learning-based methods. However, higher ENL values do not always imply superior denoising performance. Instead, abnormally high ENL values of PPB and SAR-BM3D often indicate oversmoothing in their denoising results. Among the deep learning-based methods, MRDDANet slightly outperforms in terms of ENL, while B2UB demonstrates inferior performance. Furthermore, our proposed SARDeSeg outperforms other deep learning-based methods regarding the ENL metric.

Second, we examine FOM and RGPI, with larger values indicating better preservation of detailed edge features. Compared to deep learning-based methods, traditional PPB and SAR-BM3D show apparent disadvantages, especially for PPB. Our proposed SARDeSeg achieves the best FOM performance among the deep learning-based methods, followed by NB2NB and MRDDANet. For the RGPI, B2UB exhibits the best performance among deep learning-based methods, with SARDeSeg and SAR-RDCP following in the second and third places, respectively. Although B2UB achieves the highest RGPI value, it comes at the cost of significantly poorer speckle noise smoothing capabilities, as indicated by its lower ENL values. In contrast, our proposed SARDeSeg exhibits comparable RGPI values (slightly lower than B2UB by 0.03) with significantly superior speckle noise smoothing.

Finally, we check MoR, whose value closer to 1 indicates better radiometric protection. The supervised deep learning-based methods (i.e., SAR-DRN, SAR-RDCP, MONet, and MRDDANet) exhibit MoR values that lag behind other methods. This is due to the difference between their training and test data. The former are simulated noisy images obtained from optical images, while the latter are real SAR images. In addition, methods using real SAR images as training data, particularly B2UB and our proposed SARDeSeg, exhibit the best and second-best performance, respectively.

To provide a more reliable and objective evaluation, mitigating evaluation bias resulting from a limited number of test cases, we present an average quantitative comparison of objective indicators obtained from 100 test cases in Table III. The traditional PPB exhibits oversmoothing, i.e., an excessive ability to smooth speckle noise and insufficient edge protection. Supervised deep learning methods (i.e., SAR-DRN, SAR-RDCP, MONet, and MRDDANet) demonstrate moderate speckle noise smoothing and edge protection abilities and weak radiometric protection performance. Self-supervised deep learning methods (i.e., NB2NB and B2UB) exhibit slightly better radiometric protection performance but struggle to balance speckle noise smoothing and detailed feature protection. In contrast, our proposed SARDeSeg offers more comprehensive and excellent performance. In particular, SARDeSeg has greater speckle noise smoothing ability

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TABLE III

AVERAGE QUANTITATIVE COMPARISON OF DENOISING PERFORMANCE ON 100 TEST IMAGES. UNDERLINE INDICATES THE BETTER ONE OF THE TRADITIONAL METHODS. BOLD INDICATES THE BETTER ONE OF THE DEEP LEARNING-BASED METHODS

Method	ENL [11]	FOM [25]	RGPI [45]	MoR [46]
PPB [10]	<u>283.1847</u>	0.9286	-0.2389	0.9588
SAR-BM3D [11]	210.8625	<u>0.9376</u>	<u>0.0057</u>	<u>0.9656</u>
SAR-DRN [21]	85.1403	0.9392	0.0625	0.9102
SAR-RDCP [24]	69.9250	0.9458	0.0790	0.9065
MONet [27]	82.3313	0.9382	0.0676	0.9210
MRDDANet [12]	82.0008	0.9496	0.0796	0.9203
NB2NB [13]	87.1317	0.9541	0.0507	0.9656
B2UB [34]	38.2396	0.9547	**0.0871**	**0.9735**
SARDeSeg	**133.1855**	**0.9562**	0.0870	0.9694



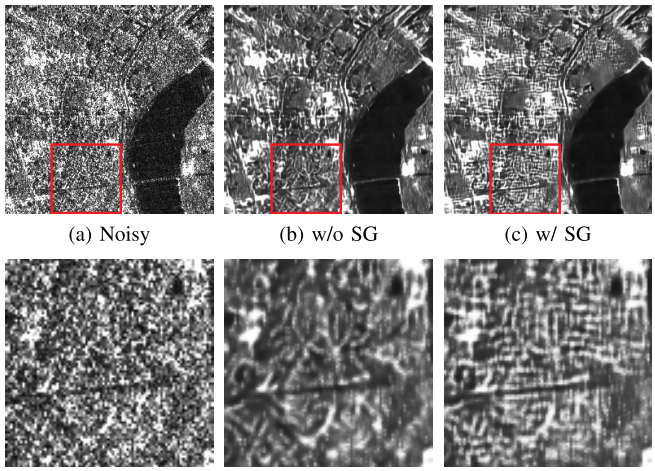
(a) Noisy (b) w/o WT-Connect (c) w/ WT-Connect

Fig. 13. Visual comparison of denoising results for the Yunnan region image using w/o WT-Connect and w/ WT-Connect.

TABLE IV

QUANTITATIVE COMPARISON OF THE EFFECT OF THE SG. BOLD INDICATES THE BETTER ONE

Configuration	ENL [11]	FOM [25]	RGPI [45]	MoR [46]
w/o SG	**64.9748**	0.9489	0.0166	0.9561
w/ SG	60.8921	**0.9630**	**0.1964**	**0.9672**



(a) Noisy (b) w/o SG (c) w/ SG

(d) Detail of Noisy (e) Detail of w/o SG (f) Detail of w/ SG

Fig. 14. Visual comparison of denoising results for the Guangdong region image using w/o SG and w/ SG.

than other deep learning-based methods, as evidenced by its impressive improvement of nearly 50 over NB2NB in the average ENL value. In addition, SARDeSeg provides better edge protection than other methods, as indicated by its improvement of around 0.02 over MONet in the average FOM value and its improvement of roughly 0.03 over NB2NB in the average RGPI value. Moreover, SARDeSeg exhibits better radiometric protection ability than supervised deep learning-based methods, improving about 0.04 over MRDDANet in the average MoR value.

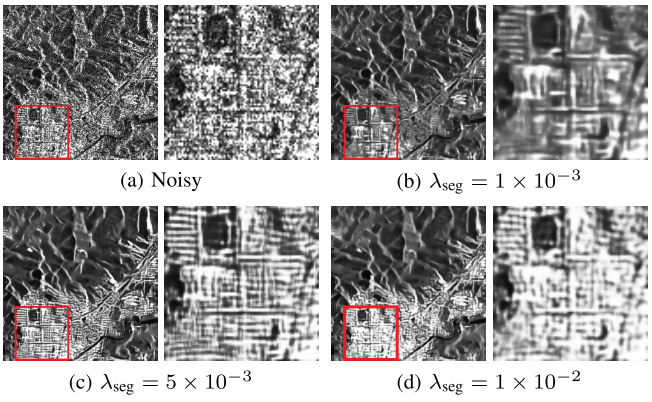
D. Ablation Studies

We conduct a series of ablation studies to illustrate the effectiveness of several key designs in our proposed SARDeSeg.

1) Effect of WT-Connect: To investigate the role of the WT-Connect in propagating semantic information provided by the segmentation network and improving denoising performance, we conduct an ablation experiment without and with the WT-Connect, denoted as w/o WT-Connect and w/WT-Connect, respectively. It is noticeable from Fig. 13 that the denoising result cannot correctly reflect the actual visual effect when the wavelet-transform-based connector is not employed. We analyze that the transmission of semantic information is difficult in the simple image space domain, and effective transmission requires decomposing the features of different frequencies in the frequency domain space. This shows that the WT-Connect is necessary for transmitting semantic information effectively in our proposed SARDeSeg, rather than being an optional means to improve the denoising performance.

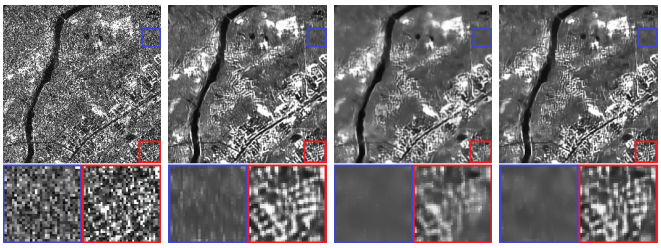
2) Effect of SG: To investigate the effect of SG on the denoising performance of our proposed SARDeSeg, we conduct an ablation study, including SARDeSeg without and with SG, denoted as w/o SG and w/ SG, respectively. Fig. 14 and Table IV compare the visual effects and quantitative metrics, respectively. The results demonstrate that incorporating SG enhances the depiction of detailed edge features, as shown in Fig. 14. This observation is also confirmed by the higher values of the FOM and RGPI metrics shown in Table IV, which indicates the more complete preservation of details and edges, respectively. By comparing whether or not SG is used, it can be seen that the denoising network alone may not be able to achieve complete preservation of details and edges, but this can be achieved through the semantic guidance using the segmentation network, followed by the transmission to the denoising network by using WT-Connect. In addition, we conduct a comparative experiment to illustrate the rationale behind setting the weight of the segmentation loss λ_{seg} in (12) to 5×10^{-3} in our proposed SARDeSeg. We set the weight λ_{seg} to 1×10^{-3}, 5×10^{-3}, and 1×10^{-2} and compare the corresponding denoising results, as presented in Fig. 15. The results reveal that setting the segmentation loss weight to a small value ($\lambda_{\text{seg}} = 1 \times 10^{-3}$) makes it challenging to observe fine textures. Conversely, when the weight is increased to a large value ($\lambda_{\text{seg}} = 1 \times 10^{-2}$), the detailed textures appear unnaturally exaggerated. The setting used in our proposed SARDeSeg ($\lambda_{\text{seg}} = 5 \times 10^{-3}$) avoids excessive exaggeration of detailed textures and ensures their clear visibility. These results indicate that SG positively affects edge feature preservation while maintaining comparable speckle suppression.

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(a) Noisy (b) $\lambda_{\text{seg}} = 1 \times 10^{-3}$ (c) $\lambda_{\text{seg}} = 5 \times 10^{-3}$ (d) $\lambda_{\text{seg}} = 1 \times 10^{-2}$

Fig. 15. Visual comparison of denoising results for the Guizhou region image using $\lambda_{\text{seg}} = 1 \times 10^{-3}$, $\lambda_{\text{seg}} = 5 \times 10^{-3}$, and $\lambda_{\text{seg}} = 1 \times 10^{-2}$.



(a) Noisy (b) w/o smoothing (c) w/ TV (d) w/ edge

Fig. 16. Visual comparison of denoising results for the Anhui region image using w/o smoothing, w/ TV smoothing, and w/ edge smoothing.

TABLE V

QUANTITATIVE COMPARISON OF THE EFFECT OF DIFFERENT SMOOTHING LOSSES. BOLD AND UNDERLINE INDICATE THE BEST AND SECOND BETTER

Configuration	ENL [11]	FOM [25]	RGPI [45]	MoR [46]
w/o smoothing	38.8005	<u>0.9349</u>	<u>0.2157</u>	<u>0.9569</u>
w/ TV smoothing	**261.5689**	0.9049	0.1837	0.9325
w/ edge smoothing	<u>92.6048</u>	**0.9504**	**0.2458**	**0.9882**

3) Effect of Edge-Aware Smoothing Loss: We conduct an ablation experiment to examine the effect of edge-aware smoothing loss on the denoising performance of our proposed SARDeSeg. Specifically, we compare the denoising results obtained without smoothing loss (i.e., w/o smoothing), with TV smoothing loss $\mathcal{L}_{\text{tv}}$ (i.e., w/ TV smoothing), and with edge-aware smoothing loss $\mathcal{L}_{\text{edge}}$ (i.e., w/ edge smoothing), as shown in Fig. 16. SARDeSeg achieves a significant reduction in speckle noise and discernible detailed edge features when no smoothing loss is applied compared to the input noisy SAR image. However, residual speckle noise remains in the denoised image. In addition, it can be noted that the TV smoothing loss significantly reduces speckle noise, but the visibility of complex edge features is compromised. In contrast, our proposed edge-aware smoothing loss can effectively reduce speckle noise while preserving the visibility of detailed edge features. The effectiveness of our approach is supported by the quantitative metrics presented in Table V. Notably, the ENL value achieved with TV smoothing loss is considerably higher than that of the other approaches, but it is accompanied by lower FOM and RGPI values. In contrast, using edge-aware smoothing loss improves ENL satisfactorily without reducing or even slightly enhancing the FOM and RGPI values. Overall, the visual and quantitative comparisons demonstrate the efficacy of the edge-aware smoothing loss for enhancing speckle noise reduction and preserving detailed edge features.

TABLE VI

QUANTITATIVE COMPARISON OF THE TRANSFERABILITY TO TerraSAR-X. UNDERLINE INDICATES THE BETTER ONE OF THE TRADITIONAL METHODS. BOLD INDICATES THE BETTER ONE OF THE DEEP LEARNING-BASED METHODS

Method	ENL [11]	FOM [25]	RGPI [45]	MoR [46]
PPB [10]	<u>15.3148</u>	0.9266	-0.3300	0.8842
SAR-BM3D [11]	9.4505	<u>0.9318</u>	<u>0.0557</u>	<u>0.9122</u>
SAR-DRN [21]	8.0755	0.9506	0.2037	0.9069
SAR-RDCP [24]	7.6362	0.9527	0.2015	0.9307
MONet [27]	6.7979	0.9613	0.2611	0.9354
MRDDANet [12]	6.9500	0.9684	0.2628	0.9276
NB2NB [13]	9.7739	0.9671	0.1698	0.9486
B2UB [34]	5.9665	0.9865	0.2883	**0.9985**
SARDeSeg	**13.8381**	**0.9871**	**0.2915**	0.9672

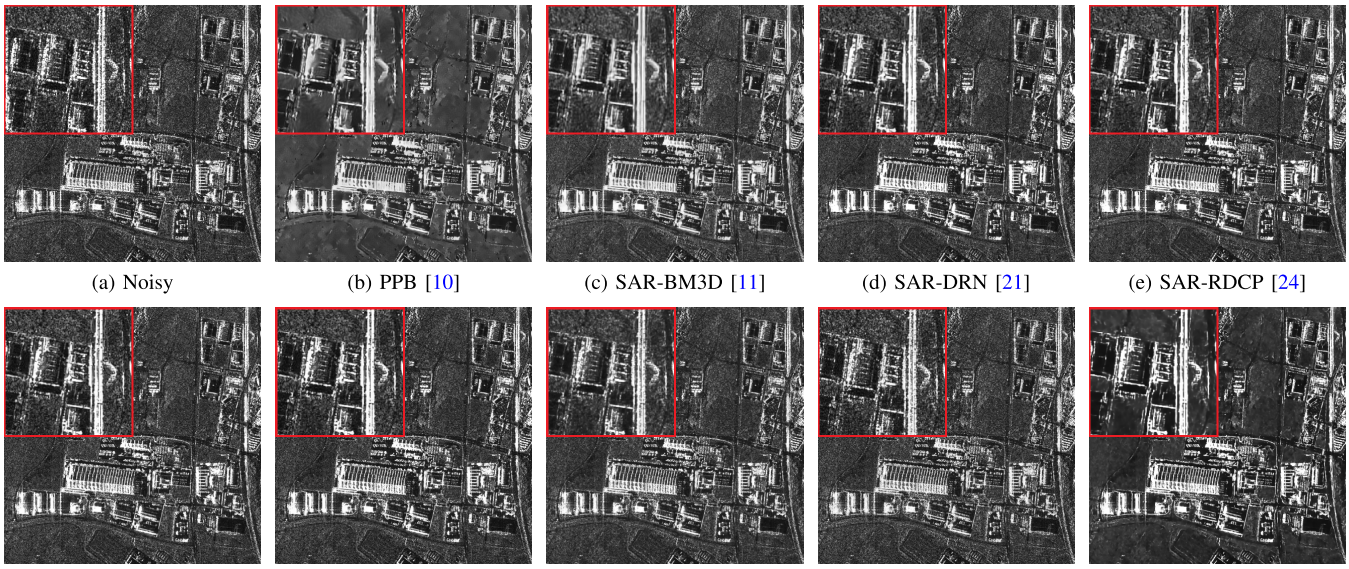
E. Transferability to Other SAR Sensors

This section presents an investigation of the transferability of our proposed SARDeSeg for denoising SAR satellite images that are not included in the training dataset. Specifically, the model is trained on Gaofen-3 images to conduct denoising tests directly on TerraSAR-X images without any fine-tuning. Fig. 17 and Table VI show visual and quantitative comparisons of denoising results for the TerraSAR-X image using various methods, respectively. Despite the fact that TerraSAR-X images are not part of the training dataset, our proposed SARDeSeg demonstrates excellent performance in reducing speckle noise, which may be due to the similarity and portability in the recognition ability of speckle noise features and semantic features from our proposed method for SAR images of different satellites. Beyond the scope of this article, potential areas of research include realizing cross-domain performance and incorporating a wider variety of real SAR images into the training dataset using self-supervised SAR image deep denoising networks.

F. Runtime Consumption and Deep Model Parameter Size

Table VII presents a comparison of the runtime and model parameter size of deep learning-based methods in the same hardware environment. In addition, the effect of different image sizes on the runtime consumed by different methods is also evaluated. It is worth noting that for our proposed SARDeSeg, once its training process is completed, the segmentation network as a guide and the WT-Connect are no longer needed, and the denoising inference of SAR images can be completed using only the trained denoising network. The results demonstrate that the deep learning-based methods require significantly less runtime than traditional methods that do not utilize deep learning. Furthermore, as the size

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(a) Noisy (b) PPB [10] (c) SAR-BM3D [11] (d) SAR-DRN [21] (e) SAR-RDCP [24]

(f) MONet [27] (g) MRDDANet [12] (h) NB2NB [13] (i) B2UB [34] (j) SARDeSeg (Ours)

Fig. 17. Visual comparison of denoising results for the TerraSAR-X image using different methods.

TABLE VII

COMPARISONS OF RUNTIME CONSUMPTION AND DEEP DENOISING MODEL PARAMETER SIZE. UNDERLINE INDICATES THE BETTER ONE OF THE TRADITIONAL METHODS. BOLD INDICATES THE BETTER ONE OF THE DEEP LEARNING-BASED METHODS

Method	Runtime (Second)			#Paras
	256×256	512×512	1024×1024	
PPB [10]	4.5598	16.7450	65.5965	-
SAR-BM3D [11]	11.6305	48.0839	197.8882	-
SAR-DRN [21]	**0.0623**	**0.0704**	**0.1042**	**0.71**M
SAR-RDCP [24]	0.0769	0.1252	0.3357	1.04M
MONet [27]	0.0709	0.0879	0.1731	2.13M
MRDDANet [12]	0.1650	0.4896	1.7849	54.50M
NB2NB [13]	0.0645	0.0766	0.1231	5.54M
B2UB [34]	0.0638	0.0745	0.1245	5.54M
SARDeSeg	0.0979	0.1459	0.4496	3.77M

of the tested image goes up, the increase in time required for deep learning-based methods is notably lower than that for traditional methods. Specifically, SAR-DRN achieves the least runtime consumption due to its lightweight deep network. MRDDANet consumes more processing time due to its more complex network architecture, thus outperforming other supervised methods (i.e., SAR-DRN, SAR-RDCP, and MONet) in SAR image denoising performance. Our proposed SARDeSeg generally achieves superior SAR image denoising performance with acceptable runtime consumption.

IV. CONCLUSION

In this article, we proposed SARDeSeg, a segmentation-guided semantic-aware self-supervised denoising method for SAR images. Our proposed SARDeSeg introduced a WT-Connect and a segmentation network jointly implemented with a self-supervised denoising network. The segmentation network serves as a guide during end-to-end training, providing the semantic information of SAR images to the self-supervised denoising network through the connector. Visual and quantitative ablation studies demonstrated the efficacy of introducing SG in our proposed SARDeSeg, resulting in more detailed and visible edge features in SAR image denoising results. Furthermore, our designed edge-aware smoothing loss effectively strengthened the suppression of speckle noise while preserving detailed features. Our proposed SARDeSeg showed competitive performance compared to SOTA methods and performed exceptionally well in preserving detailed edge features. In future work, we plan to simultaneously incorporate SAR images, including various resolutions, imaging modes, and frequencies into the training dataset, with the aim of training a unified network that can adapt to the denoising of SAR images from different sensors simultaneously.

REFERENCES

[1] Q. Chen, H. Yang, L. Li, and X. Liu, "A novel statistical texture feature for SAR building damage assessment in different polarization modes," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 154–165, 2020.

[2] Z. Cui, Q. Li, Z. Cao, and N. Liu, "Dense attention pyramid networks for multi-scale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8983–8997, Nov. 2019.

[3] M. Gong, J. Zhao, J. Liu, Q. Miao, and L. Jiao, "Change detection in synthetic aperture radar images based on deep neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 1, pp. 125–138, Jan. 2016.

[4] F. Argenti, A. Lapini, T. Bianchi, and L. Alparone, "A tutorial on speckle reduction in synthetic aperture radar images," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 3, pp. 6–35, Sep. 2013.

[5] D. Gleich and M. Datcu, "Wavelet-based despeckling of SAR images using Gauss–Markov random fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4127–4143, Dec. 2007.

[6] J. J. Ranjani and S. J. Thiruvengadam, "Dual-tree complex wavelet transform based SAR despeckling using interscale dependence," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 6, pp. 2723–2731, Jun. 2010.

[7] J. M. Bioucas-Dias and M. A. T. Figueiredo, "Multiplicative noise removal using variable splitting and constrained optimization," *IEEE Trans. Image Process.*, vol. 19, no. 7, pp. 1720–1730, Jul. 2010.

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[8] G. Chierchia, N. Pustelnik, B. Pesquet-Popescu, and J.-C. Pesquet, "A nonlocal structure tensor-based approach for multicomponent image recovery problems," *IEEE Trans. Image Process.*, vol. 23, no. 12, pp. 5531–5544, Dec. 2014.

[9] X. Ma, H. Shen, X. Zhao, and L. Zhang, "SAR image despeckling by the use of variational methods with adaptive nonlocal functionals," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3421–3435, Jun. 2016.

[10] C.-A. Deledalle, L. Denis, and F. Tupin, "Iterative weighted maximum likelihood denoising with probabilistic patch-based weights," *IEEE Trans. Image Process.*, vol. 18, no. 12, pp. 2661–2672, Dec. 2009.

[11] S. Parrilli, M. Poderico, C. V. Angelino, and L. Verdoliva, "A nonlocal SAR image denoising algorithm based on LLMMSE wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 606–616, Feb. 2012.

[12] S. Liu, Y. Lei, L. Zhang, B. Li, W. Hu, and Y.-D. Zhang, "MRDDANet: A multiscale residual dense dual attention network for SAR image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5214213.

[13] T. Huang, S. Li, X. Jia, H. Lu, and J. Liu, "Neighbor2Neighbor: A self-supervised framework for deep image denoising," *IEEE Trans. Image Process.*, vol. 31, pp. 4023–4038, 2022.

[14] G. Chen, G. Li, Y. Liu, X.-P. Zhang, and L. Zhang, "SAR image despeckling based on combination of fractional-order total variation and nonlocal low rank regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2056–2070, Mar. 2020.

[15] J. Zhang, W. Li, and Y. Li, "SAR image despeckling using multiconnection network incorporating wavelet features," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1363–1367, Aug. 2020.

[16] Y. LeCun, Y. Bengio, and G. E. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[17] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.

[18] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[19] G. Fracastoro, E. Magli, G. Poggi, G. Scarpa, D. Valsesia, and L. Verdoliva, "Deep learning methods for synthetic aperture radar image despeckling: An overview of trends and perspectives," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 2, pp. 29–51, Jun. 2021.

[20] P. Wang, H. Zhang, and V. M. Patel, "SAR image despeckling using a convolutional neural network," *IEEE Signal Process. Lett.*, vol. 24, no. 12, pp. 1763–1767, Dec. 2017.

[21] Q. Zhang, Q. Yuan, J. Li, Z. Yang, and X. Ma, "Learning a dilated residual network for SAR image despeckling," *Remote Sens.*, vol. 10, no. 2, p. 196, Jan. 2018.

[22] Y. Gui, L. Xue, and X. Li, "SAR image despeckling using a dilated densely connected network," *Remote Sens. Lett.*, vol. 9, no. 9, pp. 857–866, Sep. 2018.

[23] J. Li, Y. Li, Y. Xiao, and Y. Bai, "HDRANet: Hybrid dilated residual attention network for SAR image despeckling," *Remote Sens.*, vol. 11, no. 24, p. 2921, Dec. 2019.

[24] H. Shen, C. Zhou, J. Li, and Q. Yuan, "SAR image despeckling employing a recursive deep CNN prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 273–286, Jan. 2021.

[25] Z. Liu, R. Lai, and J. Guan, "Spatial and transform domain CNN for SAR image despeckling," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[26] J. Ko and S. Lee, "SAR image despeckling using continuous attention module," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3–19, 2022.

[27] S. Vitale, G. Ferraioli, and V. Pascazio, "Multi-objective CNN-based algorithm for SAR despeckling," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9336–9349, Nov. 2021.

[28] D. Martin, C. Fowlkes, D. Tal, and J. Malik, "A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics," in *Proc. 8th IEEE Int. Conf. Comput. Vis.*, Vancouver, BC, Canada, Jul. 2001, pp. 416–425.

[29] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, San Jose, CA, USA, Nov. 2010, pp. 270–279.

[30] J. Lehtinen et al., "Noise2Noise: Learning image restoration without clean data," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, vol. 80, Jul. 2018, pp. 2971–2980.

[31] X. Ma, C. Wang, Z. Yin, and P. Wu, "SAR image despeckling by noisy reference-based deep learning method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8807–8818, Dec. 2020.

[32] A. Krull, T.-O. Buchholz, and F. Jug, "Noise2Void—Learning denoising from single noisy images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 2124–2132.

[33] A. B. Molini, D. Valsesia, G. Fracastoro, and E. Magli, "Speckle2Void: Deep self-supervised SAR despeckling with blind-spot convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5204017.

[34] Z. Wang, J. Liu, G. Li, and H. Han, "Blind2Unblind: Self-supervised image denoising with visible blind spots," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, Jun. 2022, pp. 2017–2026.

[35] Y. Quan, M. Chen, T. Pang, and H. Ji, "Self2Self with dropout: Learning self-supervised denoising from single image," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 1887–1895.

[36] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.

[37] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.

[38] P. Liu, H. Zhang, K. Zhang, L. Lin, and W. Zuo, "Multi-level wavelet-CNN for image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Salt Lake City, UT, USA, Jun. 2018, pp. 773–782.

[39] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. NeurIPS*, Dec. 2021, pp. 12077–12090.

[40] A. G. Mullissa, D. Marcos, D. Tuia, M. Herold, and J. Reiche, "DeSpeckNet: Generalizing deep learning-based SAR image despeckling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5200315.

[41] F. Wu et al., "Built-up area mapping in China from GF-3 SAR imagery based on the framework of deep learning," *Remote Sens. Environ.*, vol. 262, Sep. 2021, Art. no. 112515.

[42] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, San Diego, CA, USA, May 2015, pp. 7–9.

[43] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, Vancouver, BC, Canada, Dec. 2019, pp. 8024–8035.

[44] S.-W. Chen, "SAR image speckle filtering with context covariance matrix formulation and similarity test," *IEEE Trans. Image Process.*, vol. 29, pp. 6641–6654, 2020.

[45] X. Ma, H. Hu, and P. Wu, "A no-reference edge-preservation assessment index for SAR image filters under a Bayesian framework based on the ratio gradient," *Remote Sens.*, vol. 14, no. 4, p. 856, Feb. 2022.

[46] X. Ma, P. Wu, Y. Wu, and H. Shen, "A review on recent developments in fully polarimetric SAR image despeckling," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 743–758, Mar. 2018.



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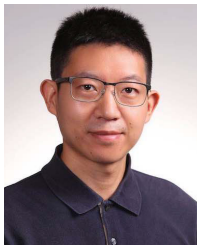
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